Q6:-

For what values of $p$ and $q$ the roots of quadratic equation $x^2 + (2p-4)x - (3q+5) = 0$ vanish?

∴ so $b=0$ and $c=0$ but we can not use $a=0$ because the answers will in undefined

∴ so as per given question:-

$a = 1$

$b = (2p-4)$

$c = -(3q+5)$

∴ so we will not use here discriminant, we have to just find out the values of $p$ and $q$ in which quadratic eq will become $= 0$

$b = 0 \rightarrow$ eq (i)

$b = (2p-4) \rightarrow$ eq (ii)

∴ so compare eq (i) with (two)

$2p - 4 = 0$

$p = \frac{4}{2}$

$\boxed{p = 2}$

Now consider a $c = 0$

$-(3q+5) = 0$

$-3q - 5 = 0$

$-3q = 5$

$q = -\frac{5}{3}$

Now prove it:-

$x^2 (4-4)x - (-5+5) = 0$

$x^2 + 0 - 0 = 0$

$x^2 = 0$

Taking root on b S.

$\sqrt{x^2} = \sqrt{0}$

$x = 0$

∴ so hence it is proved that answer are

$\text{or} = 0$